

CHERSKY, Russian Federation

WMO: 251230

Lat: 68.7451N

Lon: 161.3380E

Elev: 28

StdP: 100.99

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.8	(c) -42.0	(d) -45.5	(e) 0.0	(f) -43.4	(g) -44.0	(h) 0.1	(i) -41.7	(j) 7.5	(k) -19.8	(l) 6.4	(m) -24.1	(n) 2.0	(o) 130	(p) 0.843

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.9	(c) 25.6	(d) 17.0	(e) 23.3	(f) 15.8	(g) 21.0	(h) 14.4	(i) 17.7	(j) 23.6	(k) 16.4	(l) 22.1	(m) 15.1	(n) 20.1	(o) 3.1	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.4	10.9	20.2	13.8	9.9	18.7	12.5	9.0	17.8	50.3	23.5	46.3	22.3	42.5	20.2	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 6.5	(c) 5.7	(e) -44.9	(f) 28.8	(g) 2.5	(h) 2.2	(i) -46.7	(j) 30.4	(k) -48.2	(l) 31.7	(m) -49.6	(n) 33.0	(o) -51.4	(p) 34.6		
(a) 7.5	(b) 6.5	(c) 5.7	(e) -44.6	(f) 20.2	(g) 2.0	(h) 1.9	(i) -46.1	(j) 21.6	(k) -47.2	(l) 22.7	(m) -48.4	(n) 23.8	(o) -49.8	(p) 25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.5	-31.8	-30.6	-21.8	-12.2	0.5	10.5	12.7	10.4	4.0	-7.8	-19.9	-29.0		
	DBStd	17.60	7.38	7.33	8.62	7.80	6.34	5.86	5.19	4.71	4.22	5.92	8.09	8.50		
	HDD10.0	7377	1295	1138	986	665	299	64	30	54	186	551	898	1210		
	HDD18.3	10173	1553	1372	1244	915	553	241	189	249	429	809	1148	1468		
	CDD10.0	269	0	0	0	0	4	81	112	66	6	0	0	0		
	CDD18.3	23	0	0	0	0	0	7	14	3	0	0	0	0		
	CDH23.3	196	0	0	0	0	0	56	117	22	0	0	0	0		
	CDH26.7	32	0	0	0	0	0	6	24	2	0	0	0	0		
(14) Wind	WSAvg	2.7	2.2	2.1	2.5	2.6	3.1	3.4	3.2	3.0	3.0	2.5	2.3	2.3		
(15) Precipitation	PrecAvg	246	12	11	11	8	12	24	34	35	31	29	22	16		
	PrecMax	404	29	43	24	18	30	53	95	75	57	73	43	40		
	PrecMin	179	2	1	2	0	2	4	4	5	2	5	6	3		
	PrecStd	56	6	9	6	5	7	14	21	17	15	17	10	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.0	-3.8	0.8	6.2	19.2	27.6	29.6	26.0	18.4	6.5	1.4	0.8		
		MCWB	-4.0	-5.1	-1.0	2.3	10.6	16.6	19.3	18.0	12.5	4.5	-0.3	-1.2		
	2%	DB	-10.0	-9.4	-2.5	3.9	15.6	24.9	26.4	22.8	15.0	3.1	-1.5	-6.7		
		MCWB	-10.9	-10.6	-4.5	0.7	8.9	15.4	17.8	15.9	10.0	1.7	-3.1	-7.8		
	5%	DB	-17.8	-17.4	-5.9	1.9	12.6	22.5	24.1	20.5	12.7	1.1	-4.2	-12.7		
		MCWB	-18.2	-17.9	-7.6	-1.0	7.0	14.3	16.9	14.7	8.8	0.1	-5.4	-13.3		
	10%	DB	-22.0	-21.0	-8.7	-0.6	10.1	20.3	21.6	18.3	10.5	-0.5	-8.2	-17.4		
		MCWB	-22.2	-21.2	-10.0	-2.8	5.5	13.0	15.5	13.6	7.5	-1.5	-8.8	-17.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.9	-5.3	-1.2	2.6	11.1	17.7	20.7	19.2	13.2	4.6	0.0	-1.2		
		MCDB	-2.0	-3.5	0.6	5.3	17.7	25.0	27.4	25.1	17.3	6.4	1.0	0.8		
	2%	WB	-11.0	-10.6	-4.5	0.8	9.1	16.1	18.7	16.9	10.7	2.0	-2.9	-7.8		
		MCDB	-10.1	-9.4	-2.6	3.8	15.3	23.3	24.8	21.5	13.7	2.9	-1.6	-6.7		
	5%	WB	-18.3	-17.8	-7.6	-0.8	7.2	14.8	17.1	15.3	9.2	0.2	-5.4	-13.3		
		MCDB	-17.9	-17.5	-6.0	1.7	12.2	21.9	23.3	19.4	12.3	0.9	-4.3	-12.7		
	10%	WB	-22.2	-21.3	-10.1	-2.7	5.6	13.4	15.9	14.0	7.7	-1.4	-8.8	-17.7		
		MCDB	-22.0	-21.0	-8.7	-0.5	9.9	19.5	21.1	17.9	10.2	-0.7	-8.1	-17.4		
(35) Mean Daily Temperature Range	MDBR		5.4	5.6	8.0	9.2	8.3	9.2	8.9	8.1	5.8	4.8	5.8	5.5		
	5% DB	MCDBR	11.9	11.1	9.6	8.5	11.7	11.6	11.9	11.5	9.9	5.6	9.2	11.6		
		MCWBR	11.3	10.4	8.4	6.5	7.1	6.0	6.5	6.1	4.4	8.4	10.9			
	5% WB	MCDBR	11.4	11.1	9.5	8.4	11.3	11.3	11.5	10.8	8.9	5.1	9.3	11.2		
		MCWBR	10.9	10.4	8.4	6.4	7.2	6.3	7.2	6.8	5.9	4.3	8.6	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.165	0.219	0.289	0.336	0.370	0.358	0.411	0.387	0.315	0.253	0.182	N/A		
	taud		1.921	2.070	2.074	2.015	2.194	2.424	2.264	2.298	2.463	2.146	1.922	N/A		
	Ebn at Noon		84	571	718	785	809	841	767	740	723	498	50	0		
	Edh at Noon		6	44	85	123	117	96	110	95	61	40	5	0		
(44) All-Sky Solar Radiation	RadAvg		0.02	0.47	2.15	4.53	5.94	5.96	4.95	3.51	1.81	0.58	0.08	0.00		
	RadStd		0.00	0.05	0.10	0.15	0.45	0.44	0.45	0.34	0.16	0.08	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+1.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page